

Project Proposal

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1 Title: Dreamer v3 on Pyboy

I plan to build the DreamerV3 world model algorithm from scratch and test it on some gameboy games using Pyboy. My main objective would be Pokemon Red but there is a good chance the training is either too hard or too long. In that case I will focus on some easier games. The implementation of the Dreamerv3 will be completely build from scratch but I will use some code here and there of some other implementations on github. Moreover, I might apply some simplifications like not using the two-hot-encoding and I plan to make some modification to the model in an effort to maybe slightly improve it.

2 Papers References

- Hafner, D., Pasukonis, J., Ba, J., Lillicrap, T. (2023). Mastering Diverse Domains through World Models. arXiv preprint arXiv:2301.04104
- Pleines, M., Addis, D., Rubinstein, D., Zimmer, F., Preuss, M., Whidden, P. (2025). Pokémon Red via Reinforcement Learning. arXiv. <https://arxiv.org/abs/2502.19920>

3 Enviroment

- Pyboy games in particular Pokémon Red (if it is not too hard)